1. Title Page

- Project Title
- Course Name: *Applied Forecasting Algorithms*
- Group Members (Names & Roll Numbers)
- Instructor Name
- Institution Name
- Submission Date

2. Abstract

- Brief summary (150–250 words)
- Problem statement
- Methods used
- Key findings and conclusions

3. Introduction

3.1 Background and Motivation

- Context of the forecasting problem
- Why it is important and its impact
- What are the associated challenges?

3.2 Problem Statement

- Clear definition of the forecasting task

3.3 Objectives

- What the project aims to achieve

4. Data Source

- Origin of the dataset

4.2 Data Characteristics

- Variables, size, frequency (daily, monthly, etc.)

4.3 Data Pre-processing

- Missing values handling
- Normalization/scaling
- Feature engineering
- An exploratory analysis.

4. Methodology

- Forecasting horizon, Input-output structure, Train-test split , Hyperparameter tuning, Validation approach.
- Model implementations such AR, ARMA,ARIMA, SARIMA, ARIMAX, RNN ,GRU, LSTM ,TCN

- Ablation study, Tunned parameter values.
- Probabilistic forecasting models.

5 Analysis of the Numerical Results

A detailed analysis of the numerical results and conclusion. Research Gap Identified

6. Converting the forecast into Actions.

7. Future Work

8. References